Supplement Materials

Comparison of Different Neoadjuvant Treatments for Resectable Locoregional Esophageal Cancer – A Systematic Review and Network Meta-Analysis (NMA)

Supplement 1: PRISMA 2020 checklist to include when reporting a systematic review involving a network meta-analysis.

Supplement 2: The searching strategy

Supplement 3: The summary of the risk of bias assessment

Supplement 4: The network plot for treatments comparison of the outcomes except OS (the plot for OS was reported in the main test).

Supplement 5: The node-split analyses for OS and other outcomes.

Supplement 6: The Bayesian ranking results based on SUCRA (surface under the cumulative ranking) scores.

Supplement 7: The MCMC (Markov chain Monte Carlo) convergence diagnostics for OS and other outcomes.

Supplement 1: PRISMA 2020 checklist to include when reporting a systematic review involving a network meta-analysis.

a. The PRISMA 2020 checklist for Abstracts.

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	No
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	-
Registration	12	Provide the register name and registration number.	-

b. The PRISMA 2020 checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Yes
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 3-4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 4
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 4
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 4
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 4-5
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 4-5
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 5
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 5

Section and Topic	Item #	Checklist item	Location where item is reported
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 5
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 5
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 5
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 5
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 5
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 5
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 5
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 5
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 5
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 5
Study characteristics	17	Cite each included study and present its characteristics.	Page 5-6
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	-
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Page 6
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g.	Page 6

Section and Topic	Item #	Checklist item	Location where item is reported
		confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 6
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Page 6
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	-
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 6
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 7
	23b	Discuss any limitations of the evidence included in the review.	Page 8
	23c	Discuss any limitations of the review processes used.	Page 8
	23d	Discuss implications of the results for practice, policy, and future research.	Page 8
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 4
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	-
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Page 4
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 1
Competing interests	26	Declare any competing interests of review authors.	Page 1
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 1

Supplement 2: The search terms.

1. Pubmed

((esophagus tumor OR esophagus neoplasm OR esophagus cancer OR esophagus carcinoma) OR (esophageal tumor OR esophageal neoplasm OR esophageal cancer OR esophageal carcinoma) OR (oesophagus tumor OR oesophagus neoplasm OR oesophagus cancer OR oesophagus carcinoma) OR (oesophageal tumor OR oesophageal neoplasm OR oesophageal cancer OR oesophageal carcinoma) OR (gastroesophageal tumor OR gastroesophageal neoplasm OR gastroesophageal cancer OR gastroesophageal carcinoma) OR (gastro-oesophageal tumor OR gastro-oesophageal neoplasm OR gastro-oesophageal cancer OR gastro-oesophageal carcinoma) OR (esophagogastric tumor OR esophagogastric neoplasm OR esophagogastric cancer OR esophagogastric carcinoma) OR (oesophagogastric tumor OR oesophagogastric neoplasm OR oesophagogastric cancer OR oesophagogastric carcinoma) OR (gastric esophageal tumor OR gastric esophageal neoplasm OR gastric esophageal cancer OR gastric esophageal carcinoma)) AND ((neoadjuvant therapy OR neoadjuvant treatment OR chemotherapy) OR (neoadjuvant radiotherapy OR neoadjuvant radiation OR neoadjuvant irradiation OR preoperative radiotherapy OR preoperative radiation OR preoperative irradiation) OR (neoadjuvant chemotherapy OR preoperative chemotherapy) OR (neoadjuvant chemoradiotherapy OR neoadjuvant chemoradiation OR neoadjuvant chemoirradiation OR neoadjuvant radiochemotherapy OR preoperative chemoradiotherapy OR preoperative chemoradiation OR preoperative chemoirradiation OR preoperative radiochemotherapy) OR (preoperative treatment OR preoperative therapy))

text availability: Full text

article type: Randomized Controlled Trial

2. EmBase

(((((('esophagus'/exp OR esophagus) AND ('tumor'/exp OR tumor) OR 'esophagus'/exp OR esophagus) AND ('neoplasm'/exp OR neoplasm) OR 'esophagus'/exp OR esophagus) AND ('cancer'/exp OR cancer) OR 'esophagus'/exp OR esophagus) AND ('carcinoma'/exp OR carcinoma) OR ((((esophageal AND ('tumor'/exp OR tumor) OR esophageal) AND ('neoplasm'/exp OR neoplasm) OR esophageal) AND ('cancer'/exp OR cancer) OR esophageal) AND ('carcinoma'/exp OR carcinoma)) OR (((((('oesophagus'/exp OR oesophagus) AND ('tumor'/exp OR tumor) OR 'oesophagus'/exp OR oesophagus) AND ('neoplasm'/exp OR neoplasm) OR 'oesophagus'/exp OR oesophagus) AND ('cancer'/exp OR cancer) OR 'oesophagus'/exp OR oesophagus) AND ('carcinoma'/exp OR carcinoma)) OR ((((oesophageal AND ('tumor'/exp OR tumor) OR oesophageal) AND ('neoplasm'/exp OR neoplasm) OR oesophageal) AND ('cancer'/exp OR cancer) OR oesophageal) AND ('carcinoma'/exp OR carcinoma)) OR ((((gastroesophageal AND ('tumor'/exp OR tumor) OR gastroesophageal) AND ('neoplasm'/exp OR neoplasm) OR gastroesophageal) AND ('cancer'/exp OR cancer) OR gastroesophageal) AND ('carcinoma'/exp OR carcinoma)) OR ((((('gastro oesophageal' AND ('tumor'/exp OR tumor) OR 'gastro oesophageal') AND ('neoplasm'/exp OR neoplasm) OR 'gastro oesophageal') AND ('cancer'/exp OR cancer) OR 'gastro oesophageal') AND ('carcinoma'/exp OR carcinoma)) OR ((((esophagogastric

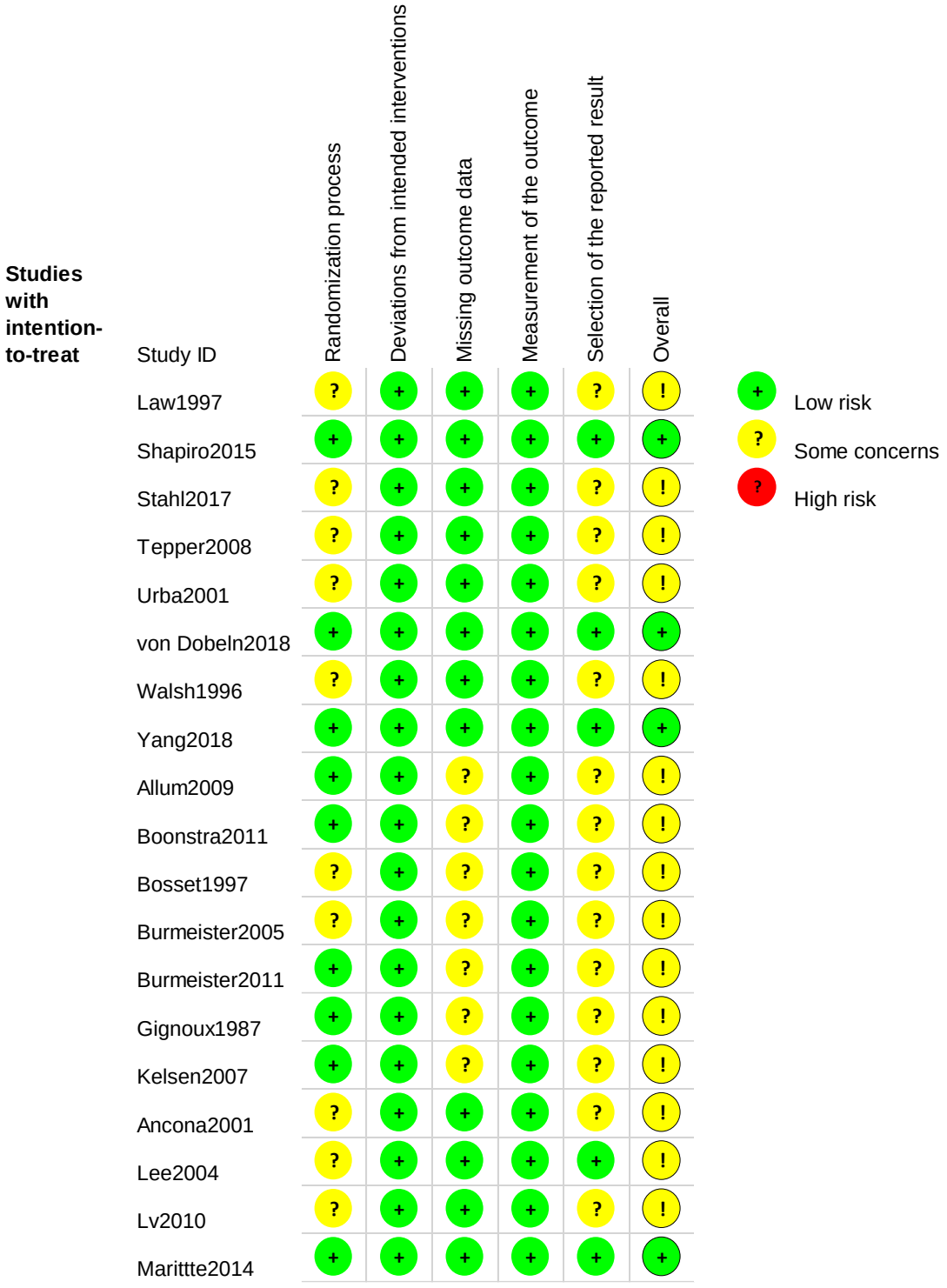
AND ('tumor'/exp OR tumor) OR esophagogastric) AND ('neoplasm'/exp OR neoplasm) OR esophagogastric) AND ('cancer'/exp OR cancer) OR esophagogastric) AND ('carcinoma'/exp OR carcinoma)) OR ((((oesophagogastric AND ('tumor'/exp OR tumor) OR oesophagogastric) AND ('neoplasm'/exp OR neoplasm) OR oesophagogastric) AND ('cancer'/exp OR cancer) OR oesophagogastric) AND ('carcinoma'/exp OR carcinoma)) OR ((((gastric AND esophageal AND ('tumor'/exp OR tumor) OR gastric) AND esophageal AND ('neoplasm'/exp OR neoplasm) OR gastric) AND esophageal AND ('cancer'/exp OR cancer) OR gastric) AND esophageal AND ('carcinoma'/exp OR carcinoma))) AND ((neoadjuvant AND ('therapy'/exp OR therapy) OR neoadjuvant) AND ('treatment'/exp OR treatment) OR 'chemotherapy'/exp OR chemotherapy OR ((((((neoadjuvant AND ('radiotherapy'/exp OR radiotherapy) OR neoadjuvant) AND ('radiation'/exp OR radiation) OR neoadjuvant) AND ('irradiation'/exp OR irradiation) OR preoperative) AND ('radiotherapy'/exp OR radiotherapy) OR preoperative) AND ('radiation'/exp OR radiation) OR preoperative) AND ('irradiation'/exp OR irradiation)) OR ((neoadjuvant AND ('chemotherapy'/exp OR chemotherapy) OR preoperative) AND ('chemotherapy'/exp OR chemotherapy)) OR (((((((neoadjuvant AND ('chemoradiotherapy'/exp OR chemoradiotherapy) OR neoadjuvant) AND ('chemoradiation'/exp OR chemoradiation) OR neoadjuvant) AND chemoirradiation OR neoadjuvant) AND ('radiochemotherapy'/exp OR radiochemotherapy) OR preoperative) AND ('chemoradiotherapy'/exp OR chemoradiotherapy) OR preoperative) AND ('chemoradiation'/exp OR chemoradiation) OR preoperative) AND chemoirradiation OR preoperative) AND ('radiochemotherapy'/exp OR radiochemotherapy)) OR ((preoperative AND ('treatment'/exp OR treatment) OR preoperative) AND ('therapy'/exp OR therapy)))
text availability: Full text
article type: Randomized Controlled Trial

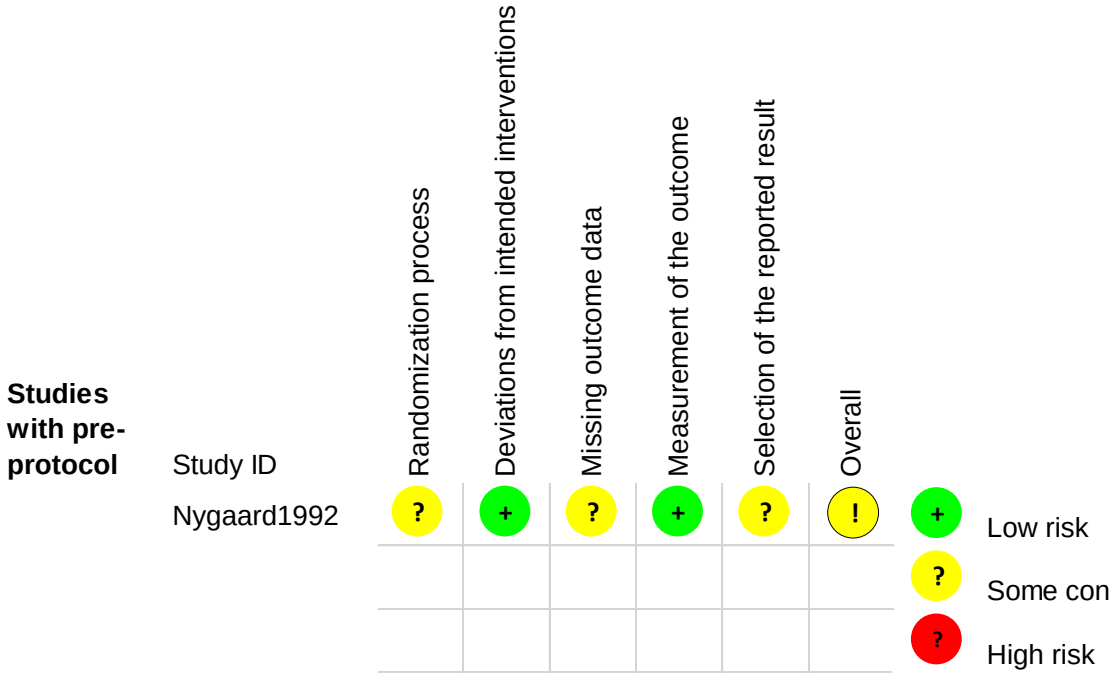
3. Cochrane Library database

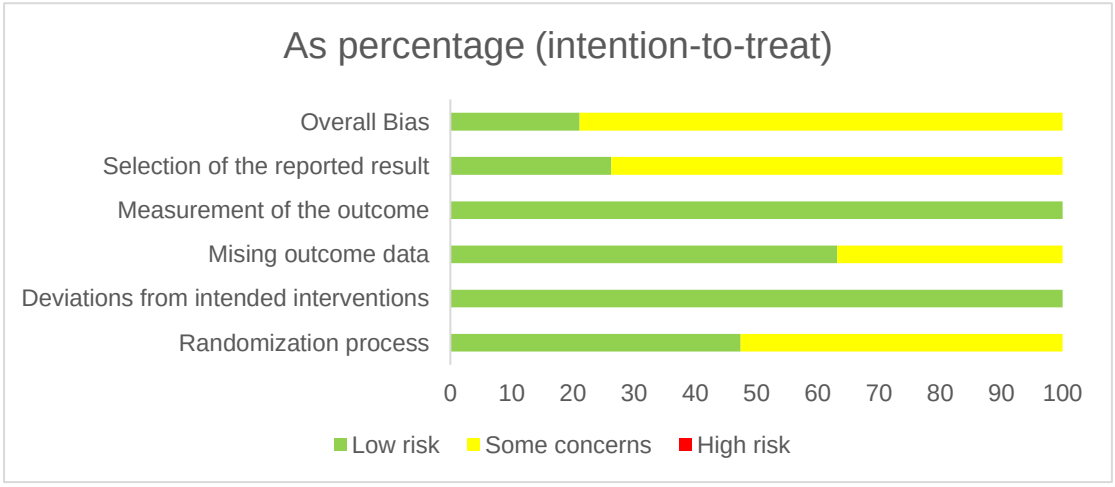
(esophagus tumor OR esophagus neoplasm OR esophagus cancer OR esophagus carcinoma) OR (esophageal tumor OR esophageal neoplasm OR esophageal cancer OR esophageal carcinoma) OR (oesophagus tumor OR oesophagus neoplasm OR oesophagus cancer OR oesophagus carcinoma) OR (oesophageal tumor OR oesophageal neoplasm OR oesophageal cancer OR oesophageal carcinoma) OR (gastroesophageal tumor OR gastroesophageal neoplasm OR gastroesophageal cancer OR gastroesophageal carcinoma) OR (gastro-oesophageal tumor OR gastro-oesophageal neoplasm OR gastro-oesophageal cancer OR gastro-oesophageal carcinoma) OR (esophagogastric tumor OR esophagogastric neoplasm OR esophagogastric cancer OR esophagogastric carcinoma) OR (oesophagogastric tumor OR oesophagogastric neoplasm OR oesophagogastric cancer OR oesophagogastric carcinoma) OR (gastric esophageal tumor OR gastric esophageal neoplasm OR gastric esophageal cancer OR gastric esophageal carcinoma)) AND ((neoadjuvant therapy OR neoadjuvant treatment OR chemotherapy) OR (neoadjuvant radiotherapy OR neoadjuvant radiation OR neoadjuvant irradiation OR preoperative radiotherapy OR preoperative radiation OR preoperative irradiation) OR (neoadjuvant chemotherapy OR preoperative chemotherapy) OR (neoadjuvant chemoradiotherapy OR neoadjuvant chemoradiation OR neoadjuvant

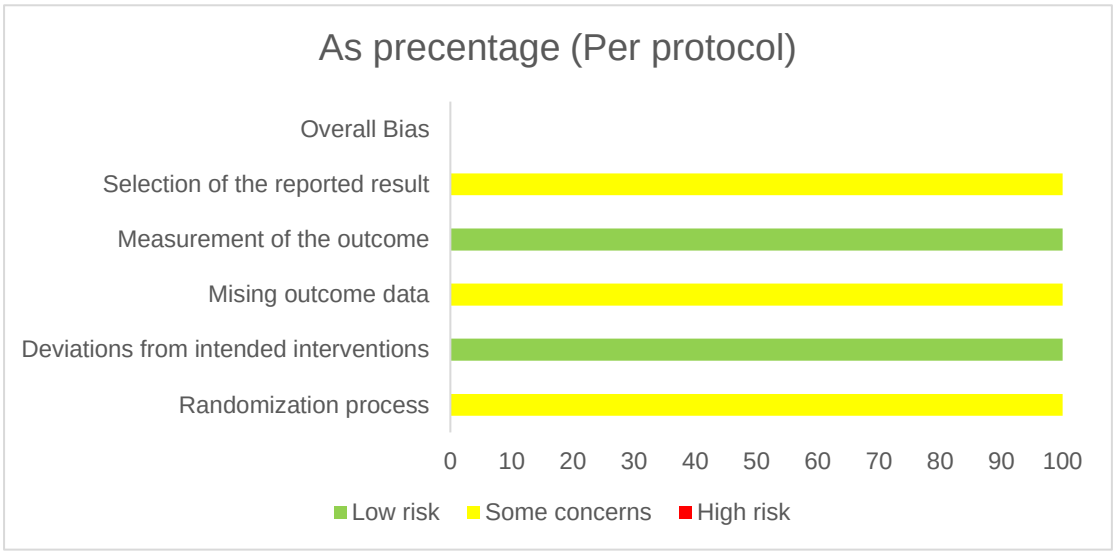
chemoirradiation OR neoadjuvant radiochemotherapy OR preoperative chemoradiotherapy OR preoperative chemoradiation OR preoperative chemoirradiation OR preoperative radiochemotherapy) OR (preoperative treatment OR preoperative therapy)) in Title Abstract Keyword - in Trials (Word variations have been searched)
Source: ICTRP AND CT.gov AND CINAHL

Supplement 3: Summary of risk of bias assessment



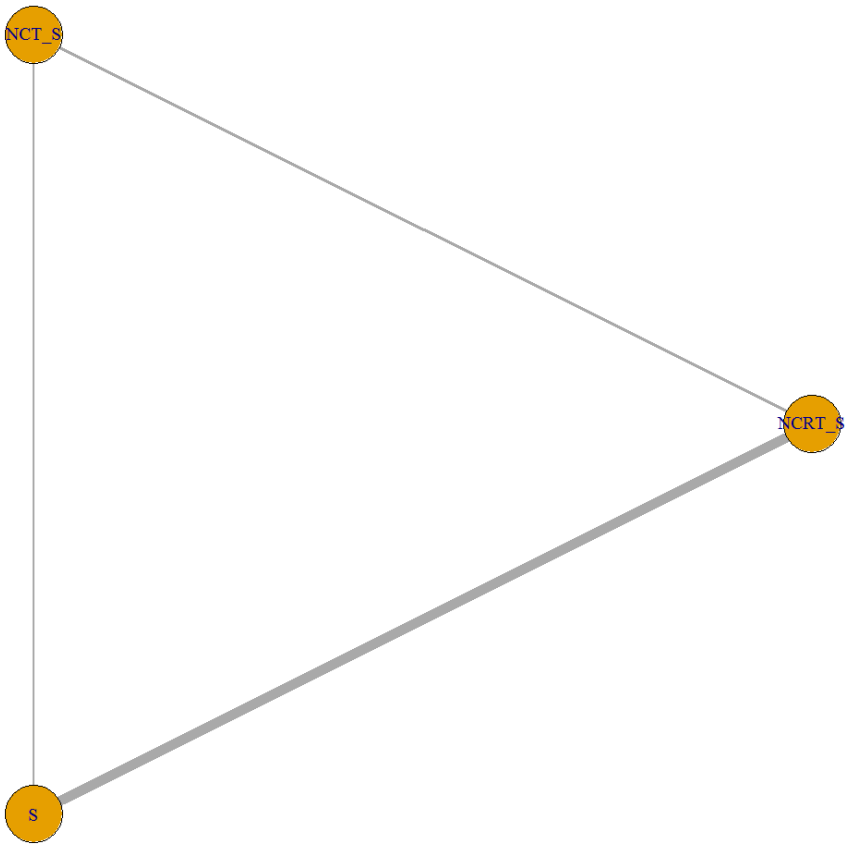

Studies with pre-protocol
Study ID
Nygaard1992
Randomization process
Deviations from intended interventions
Missing outcome data
Measurement of the outcome
Selection of the reported result
Overall
Low risk
Some con
High risk


As percentage (intention-to-treat)
Overall Bias
Selection of the reported result
Measurement of the outcome
Mising outcome data
Deviations from intended interventions
Randomization process
0
10
20
30
40
50
60
70
80
90
100
Low risk
Some concerns
High risk

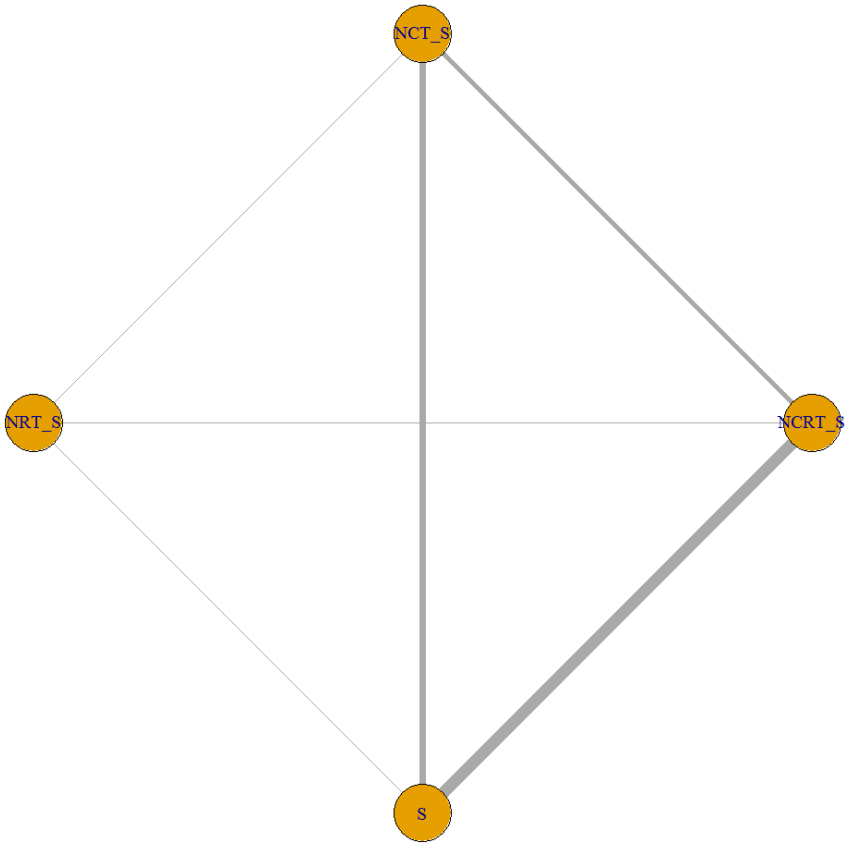

As precentage (Per protocol)
Overall Bias
Selection of the reported result
Measurement of the outcome
Mising outcome data
Deviations from intended interventions
Randomization process
0
10
20
30
40
50
60
70
80
90
100
Low risk
Some concerns
High risk

Supplement 4: The network plots for treatments comparison of PFS, PD, PC, LRR, and DM and subgroup analysis for pathological types. Each line represents a type of head-to-head comparison. The width of lines is proportional to the number of trials comparing the connected treatments. S, surgery; NCT_S, neoadjuvant chemotherapy followed by surgery; NRT_S, neoadjuvant radiotherapy followed by surgery; NCRT_S, neoadjuvant chemoradiotherapy followed by surgery.

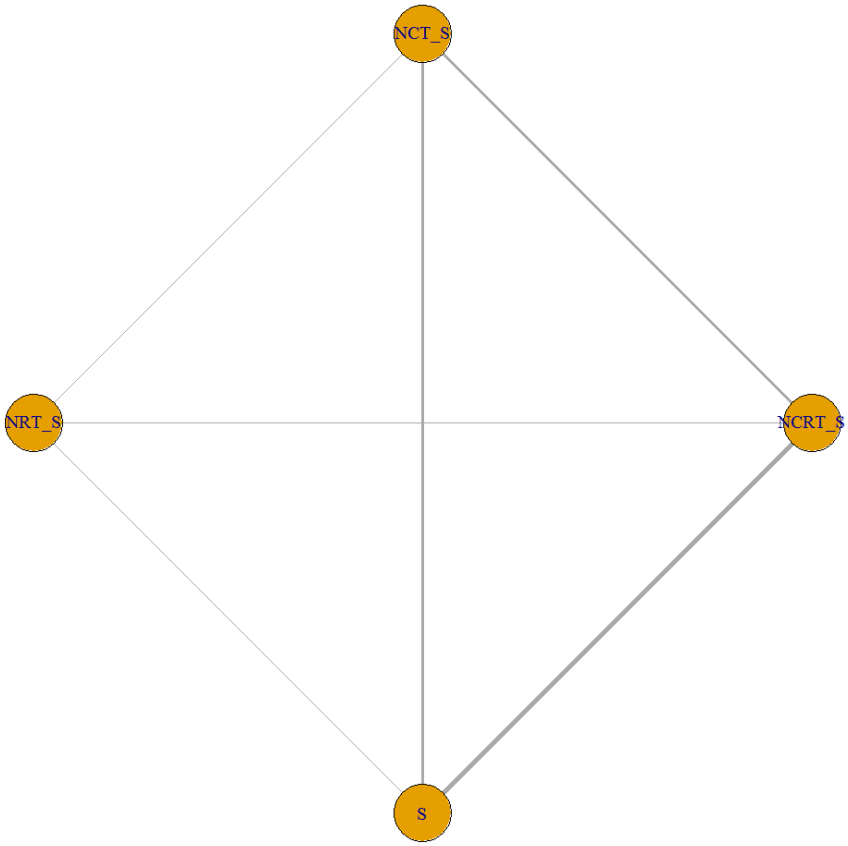
a. The network plot for treatments comparison of PFS



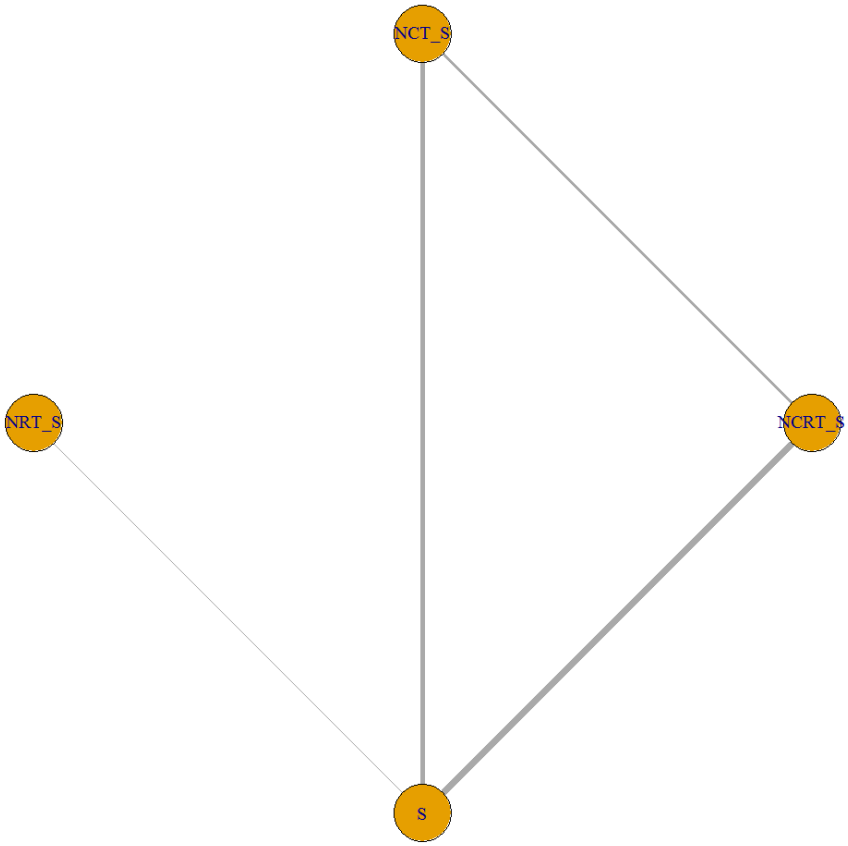
b. The network plot for treatments comparison of postoperative mortality



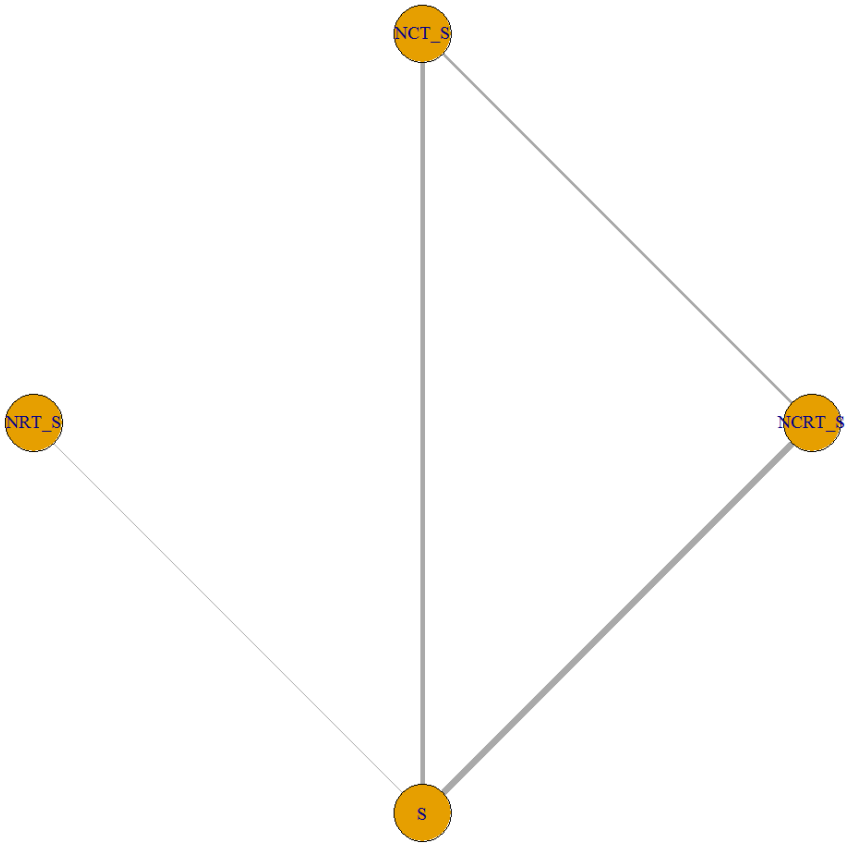
c. The network plot for treatments comparison of postoperative morbidity



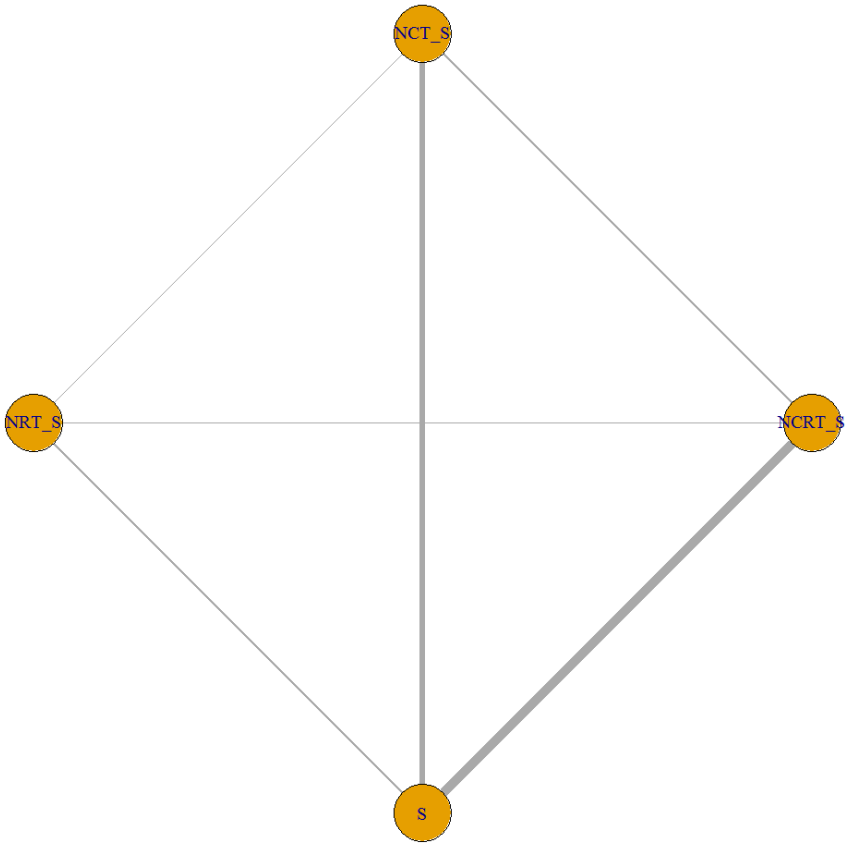
d. The network plot for treatments comparison of LRR



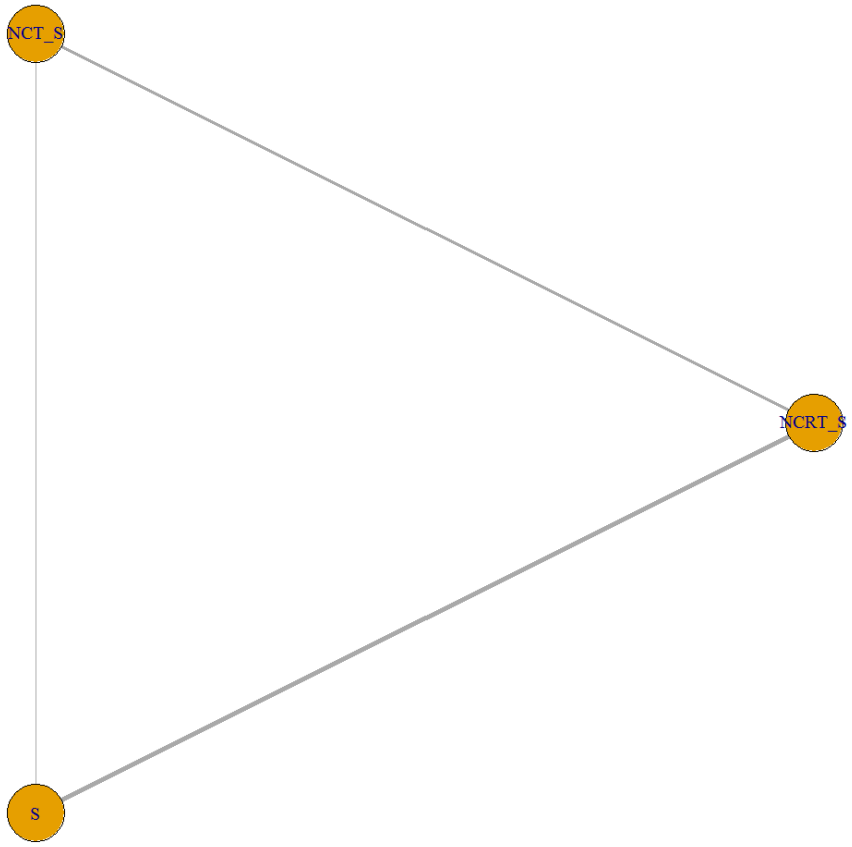
e. The network plot for treatments comparison of DM



f. Subgroup analysis for pathological types: the network plot for treatments comparison of OS for patients with squamous cell carcinoma.

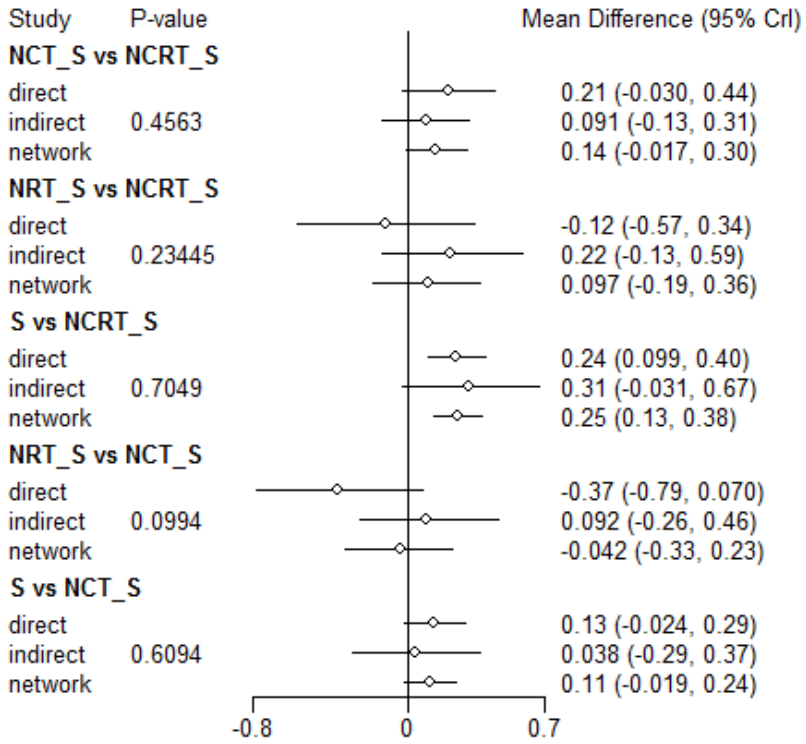


g. Subgroup analysis for pathological types: the network plot for treatments comparison of OS for patients with adenocarcinoma.

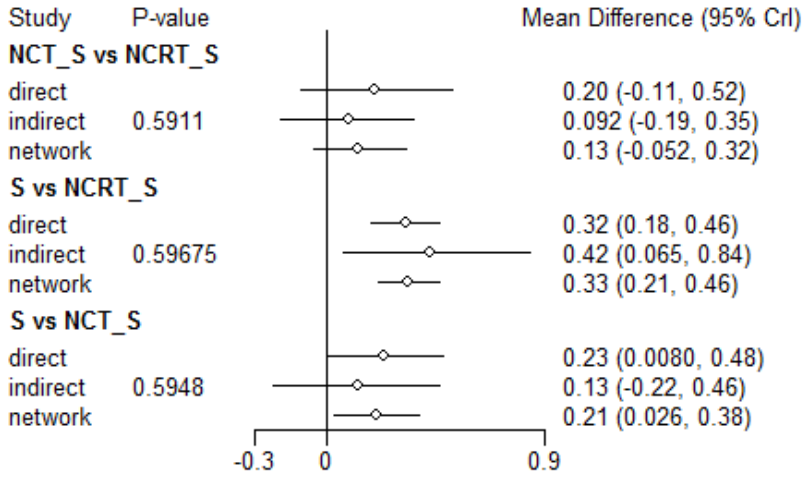


Supplement 5: The node-split analyses of inconsistency

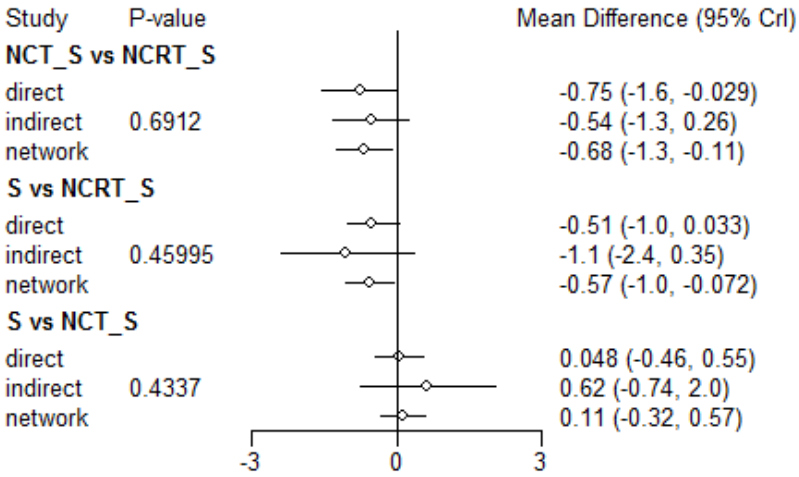
a. The node-split analysis for OS



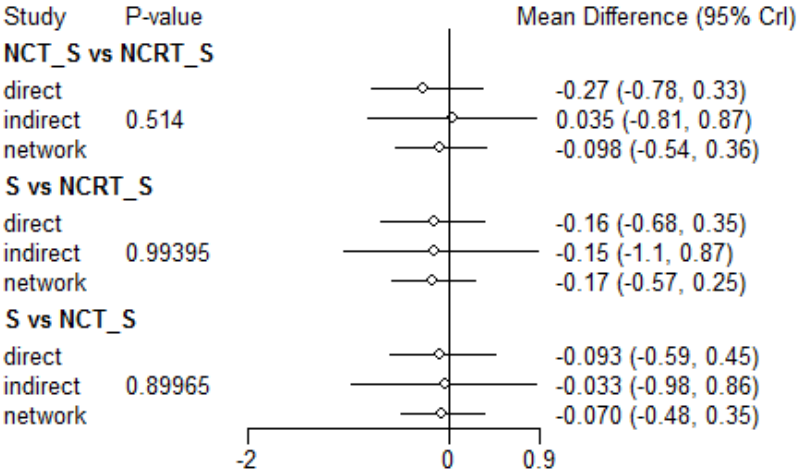
b. The node-split analysis for PFS



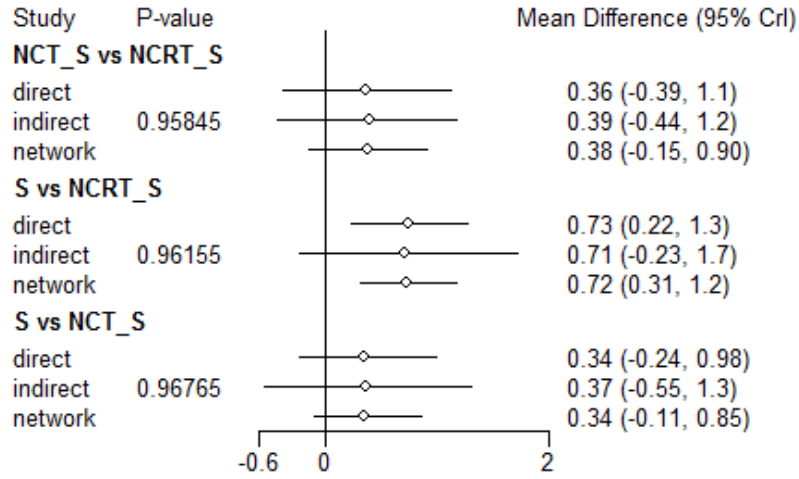
c. The node-split analysis for postoperative mortality



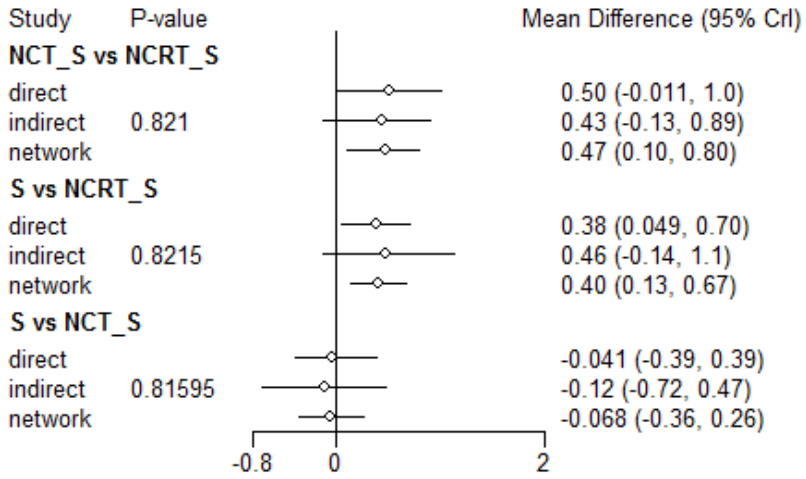
d. The node-split analysis for postoperative morbidity



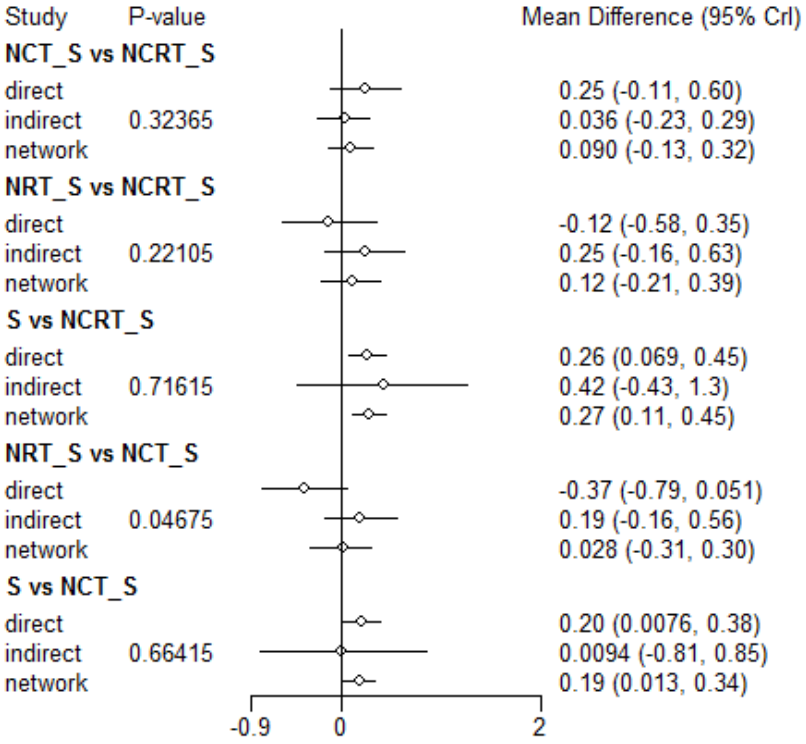
e. The node-split analysis for LRR



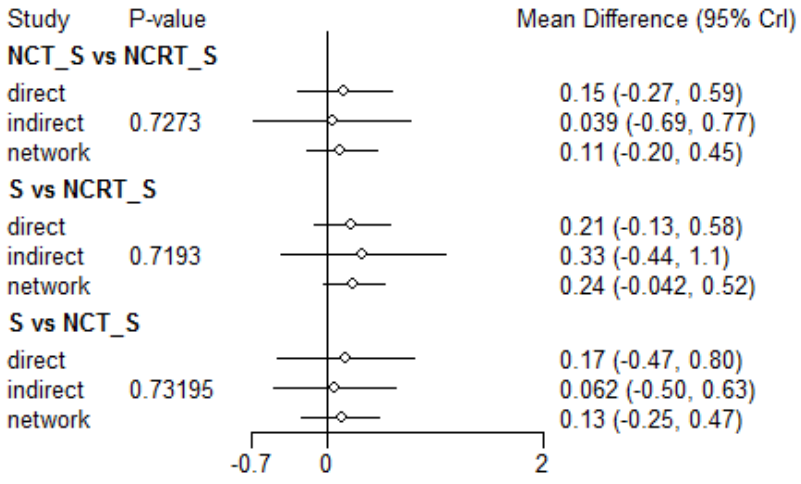
f. The node-split analysis for DM



g. Subgroup analysis for pathological types: the node-split analysis for OS for patients with squamous cell carcinoma.

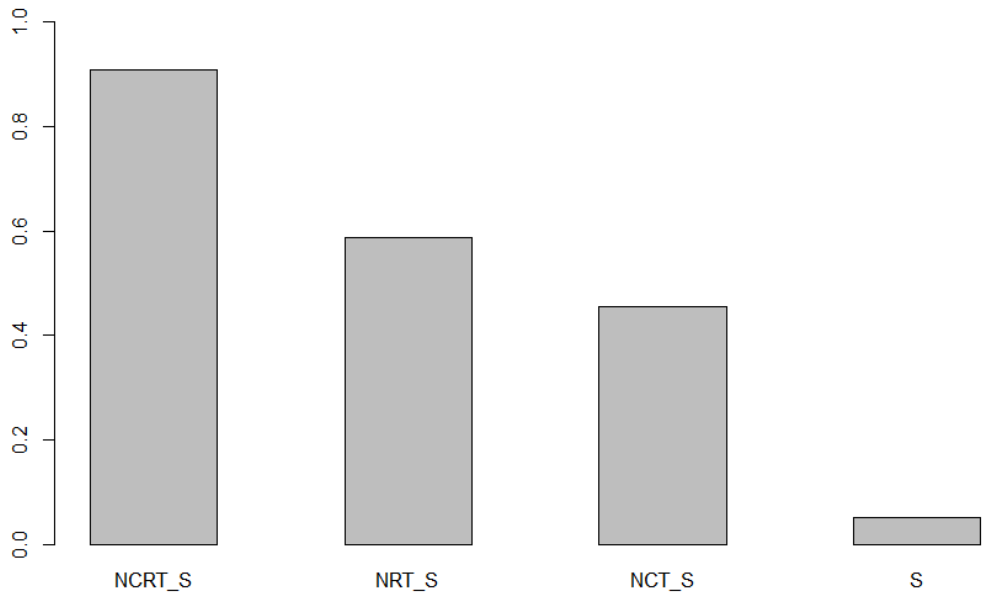


h. Subgroup analysis for pathological types: the node-split analysis for OS for patients with adenocarcinoma.

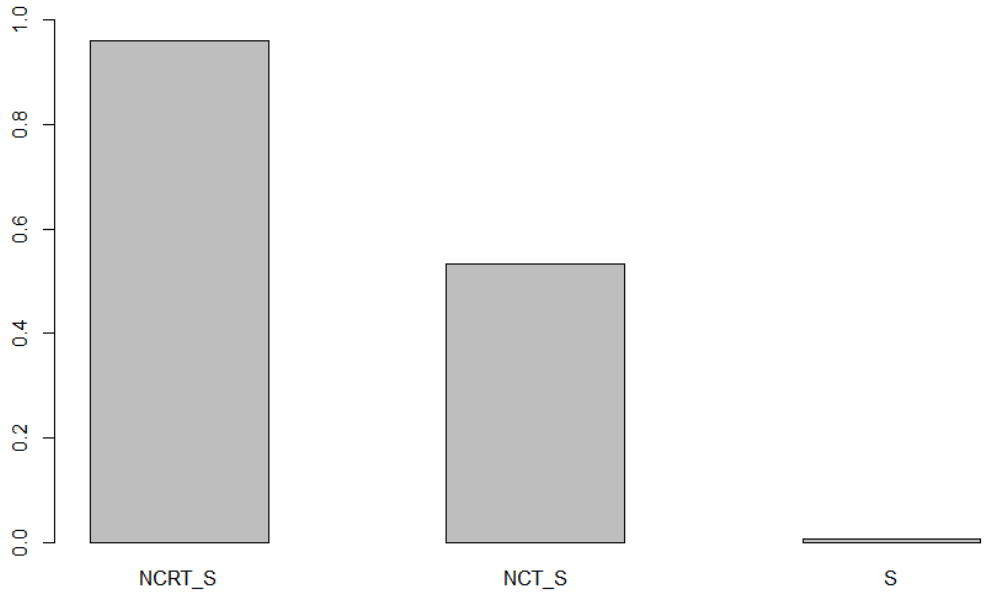

Study
P-value
Mean Difference (95% CrI)
NCT_S vs NCRT_S
direct
0.15 (-0.27, 0.59)
indirect
0.7273
0.039 (-0.69, 0.77)
network
0.11 (-0.20, 0.45)
S vs NCRT_S
direct
0.21 (-0.13, 0.58)
indirect
0.7193
0.33 (-0.44, 1.1)
network
0.24 (-0.042, 0.52)
S vs NCT_S
direct
0.17 (-0.47, 0.80)
indirect
0.73195
0.062 (-0.50, 0.63)
network
0.13 (-0.25, 0.47)
-0.7
0
2

Supplement 6: Bayesian ranking results based on SUCRA (surface under the cumulative ranking) scores.

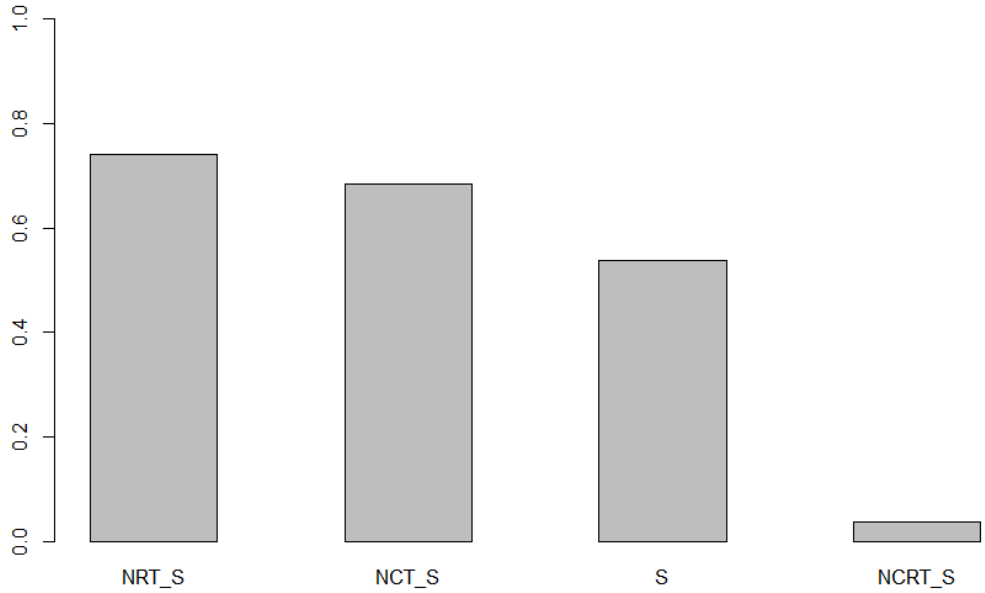
a. OS



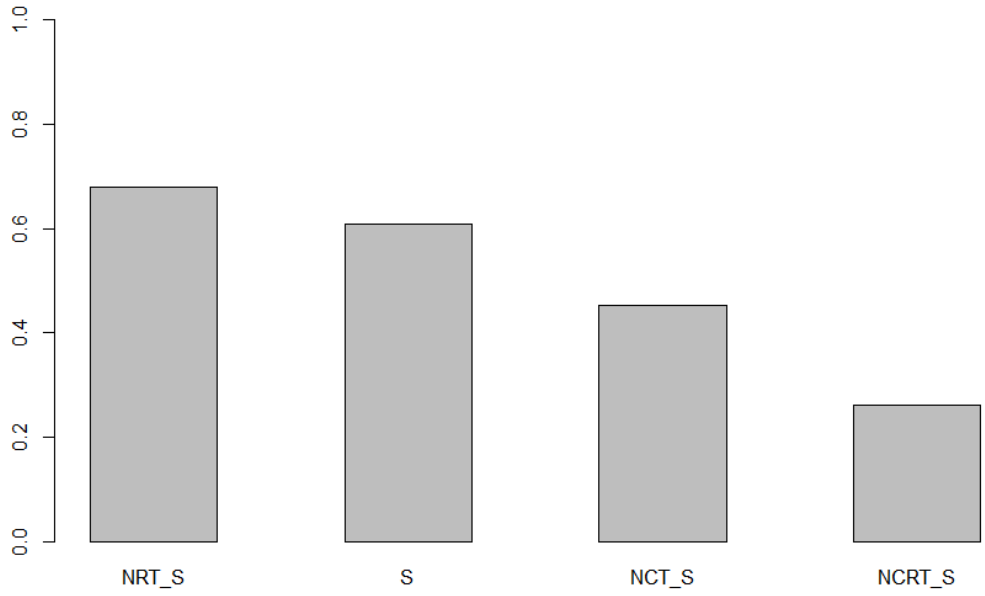
b. PFS



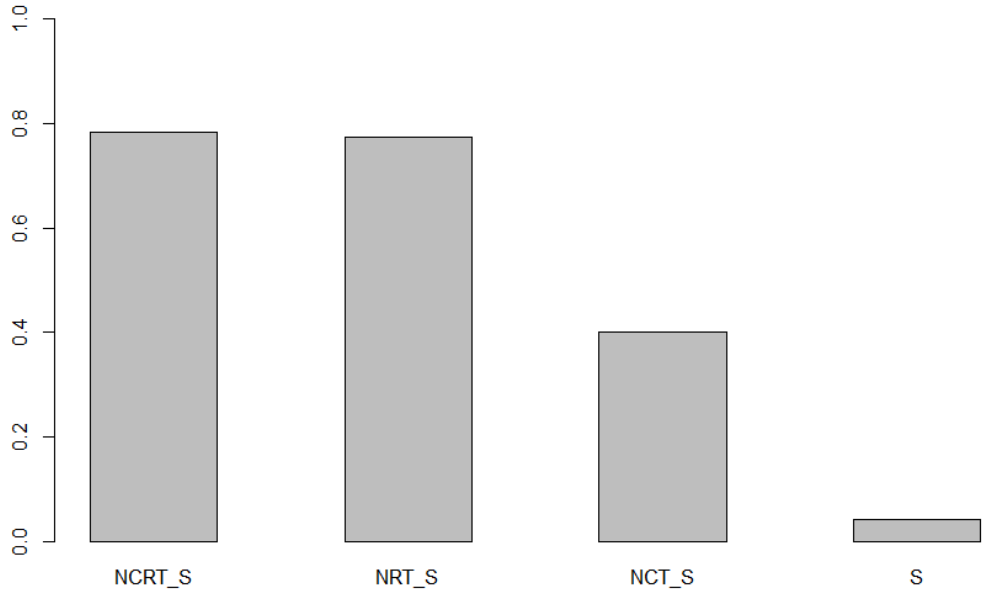
c. Postoperative mortality



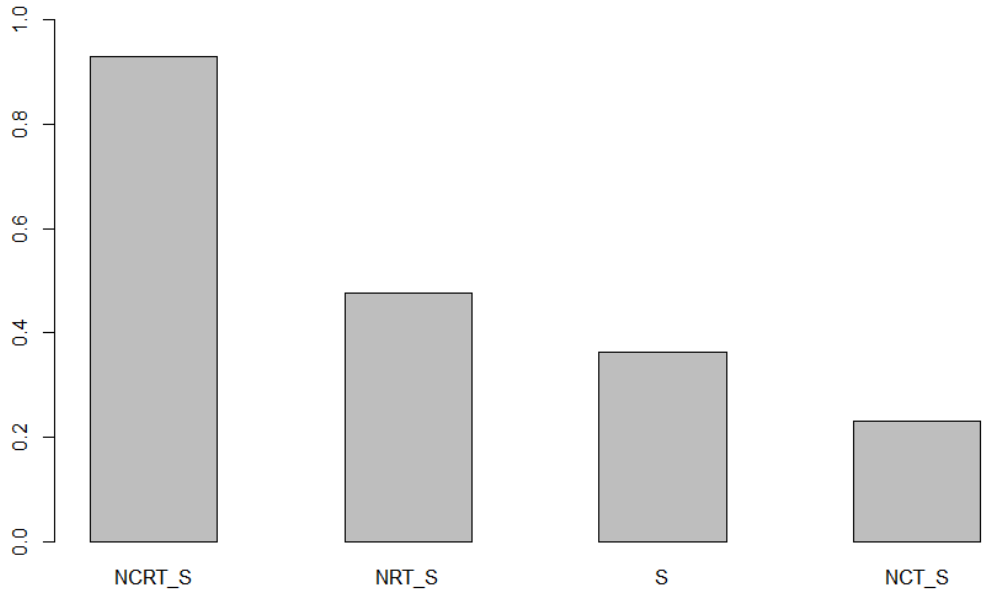
d. Postoperative morbidity



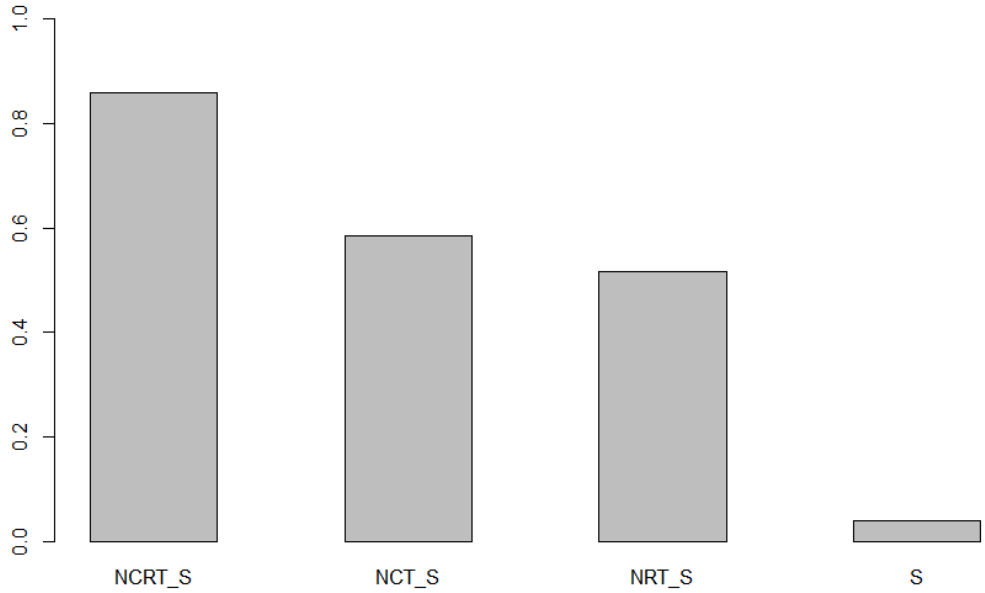
e. LRR



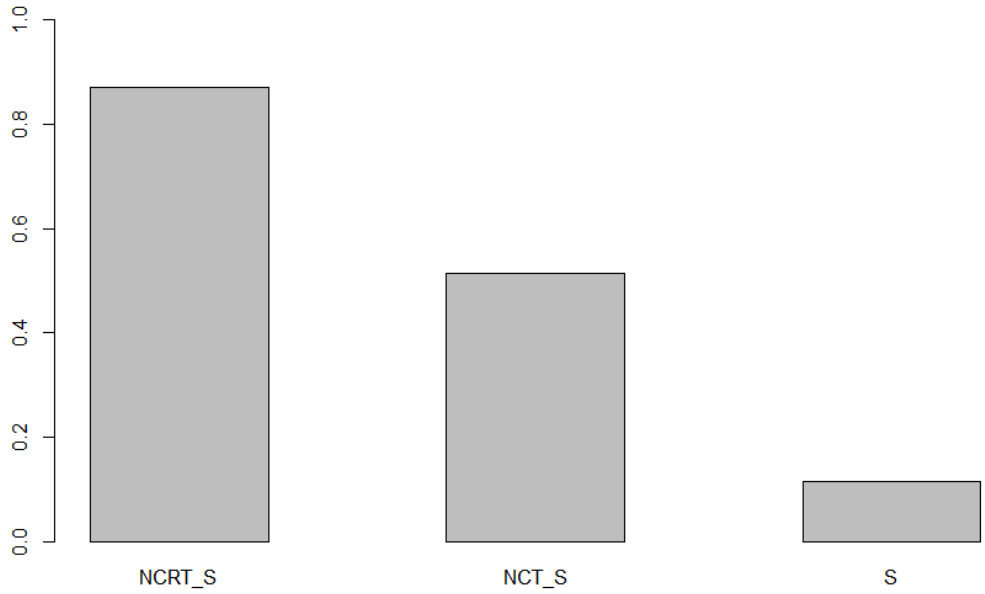
f. DM



g. Subgroup analysis for pathological types: OS for patients with squamous cell carcinoma.

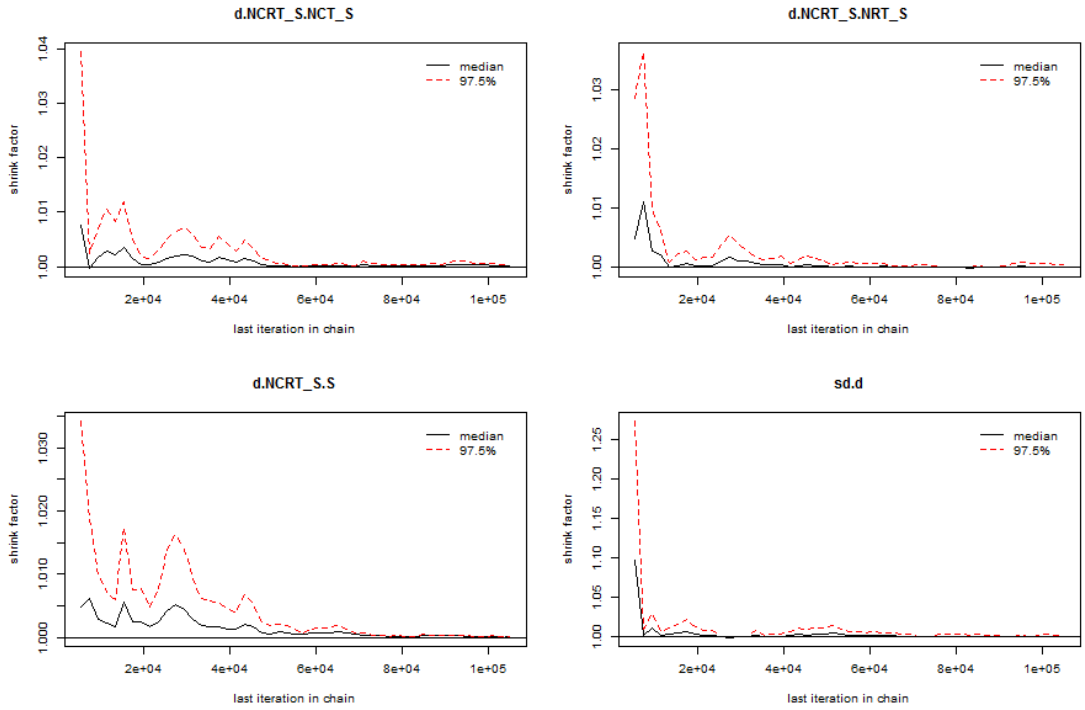


h. Subgroup analysis for pathological types: OS for patients with adenocarcinoma.

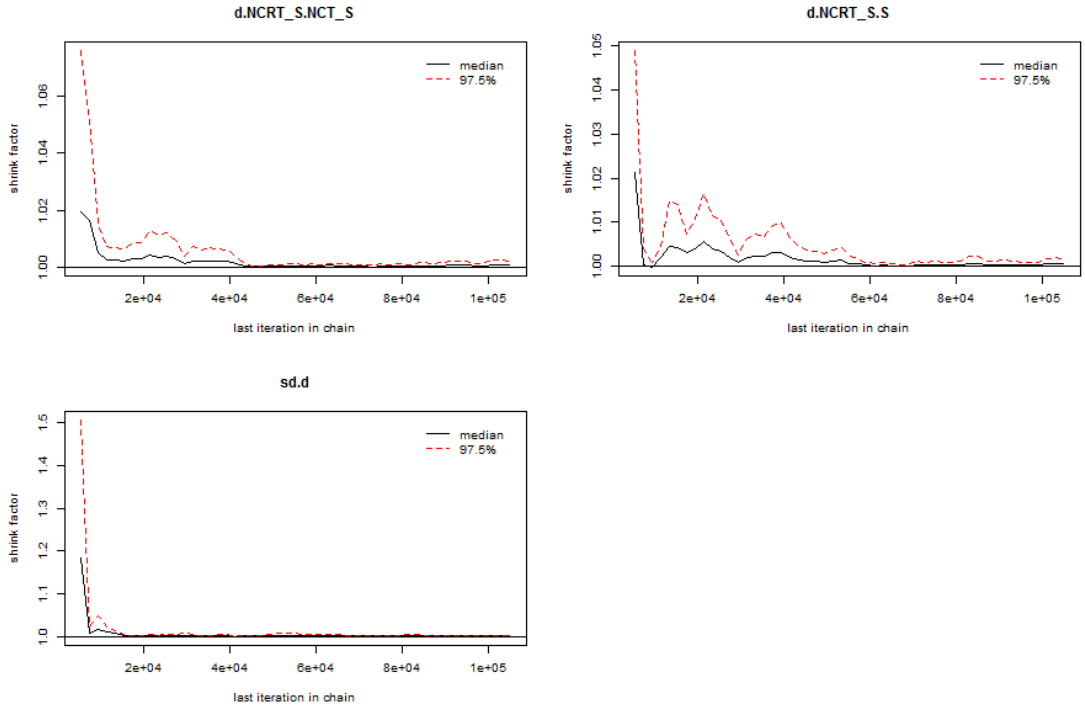


Supplement 7: The MCMC (Markov chain Monte Carlo) convergence diagnostics.

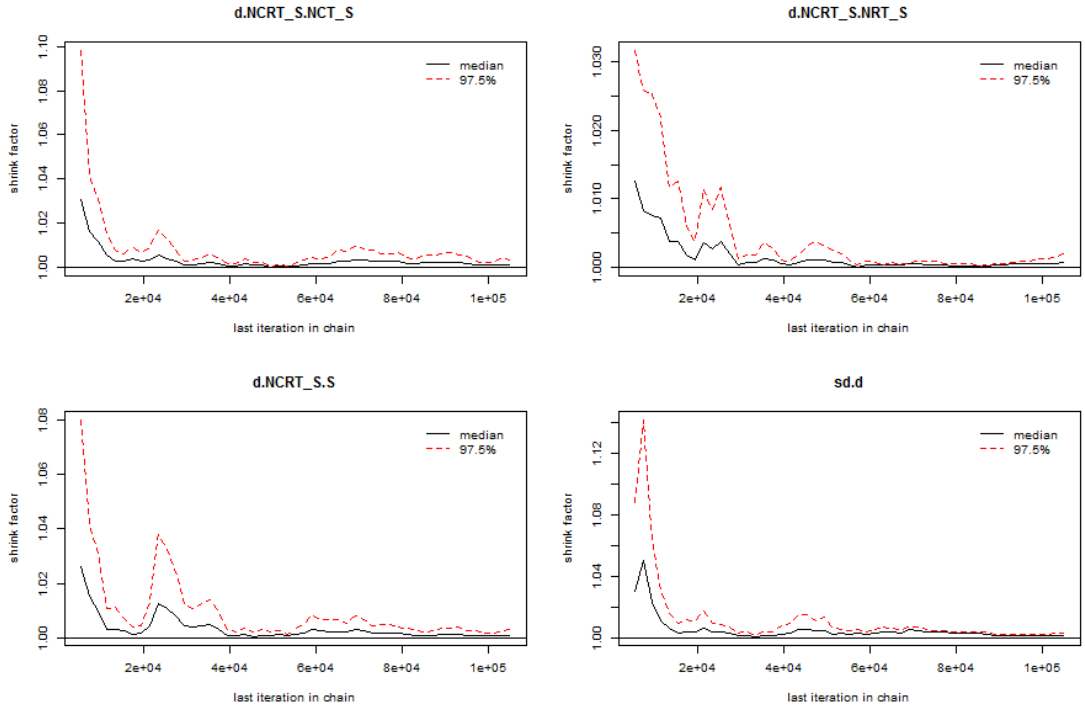
a. OS



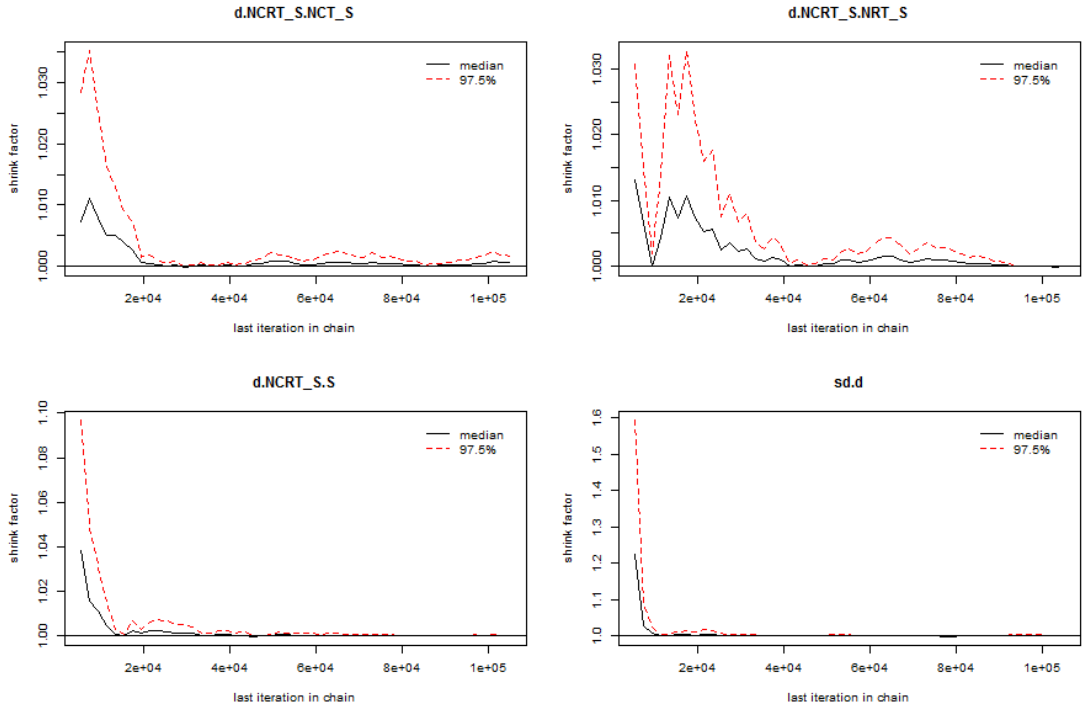
b. PFS



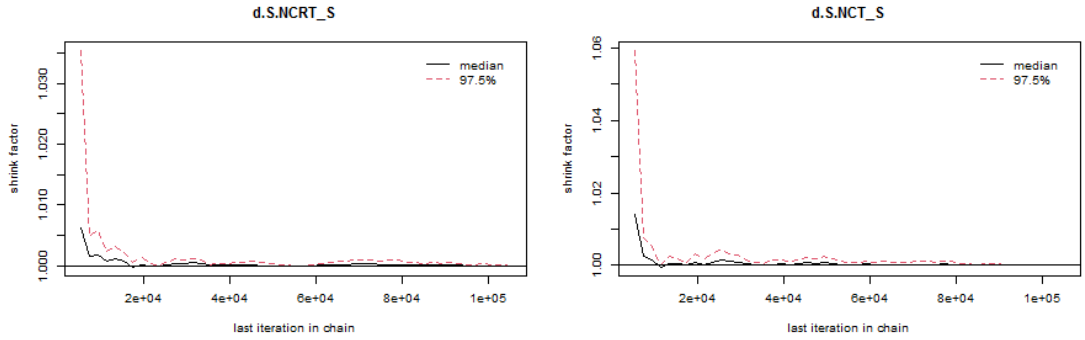
c. Postoperative mortality

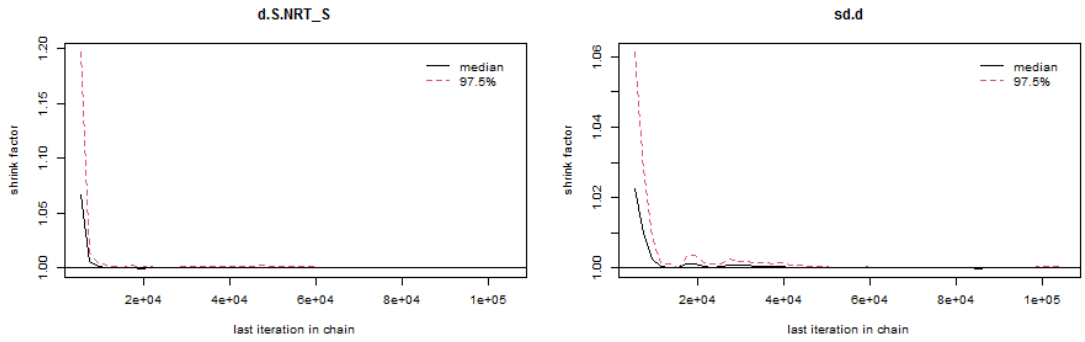


d. Postoperative morbidity

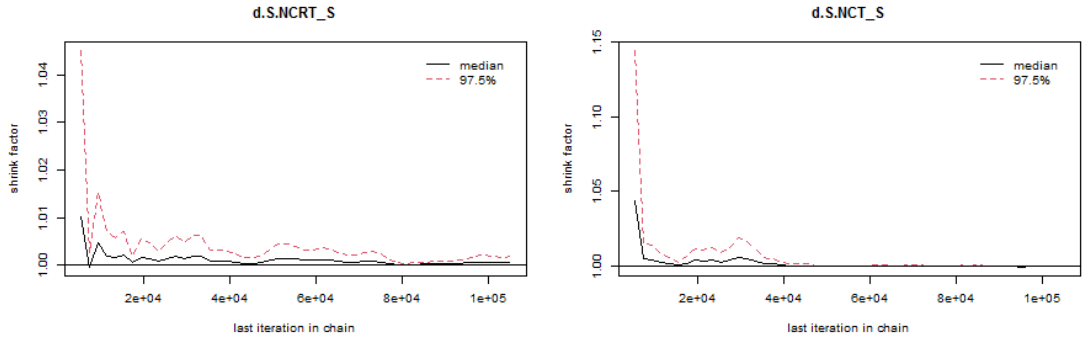


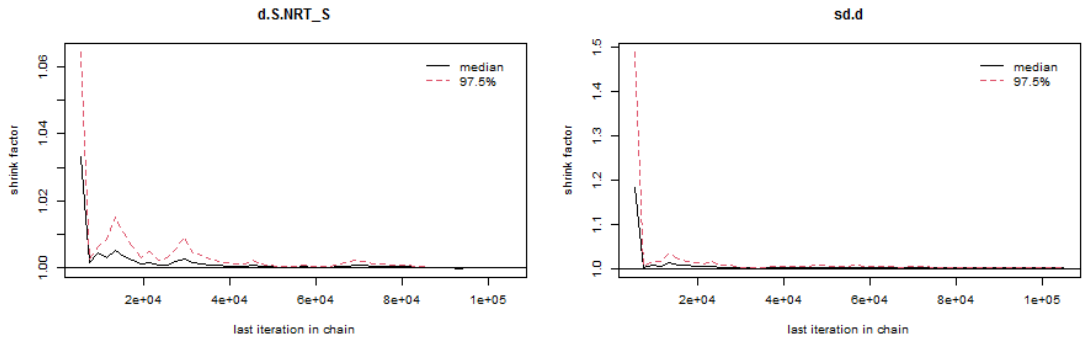
e. LRR


d.S.NCRT_S
d.S.NCT_S
median
97.5%
shrink factor
last iteration in chain
2e+04
4e+04
6e+04
8e+04
1e+05

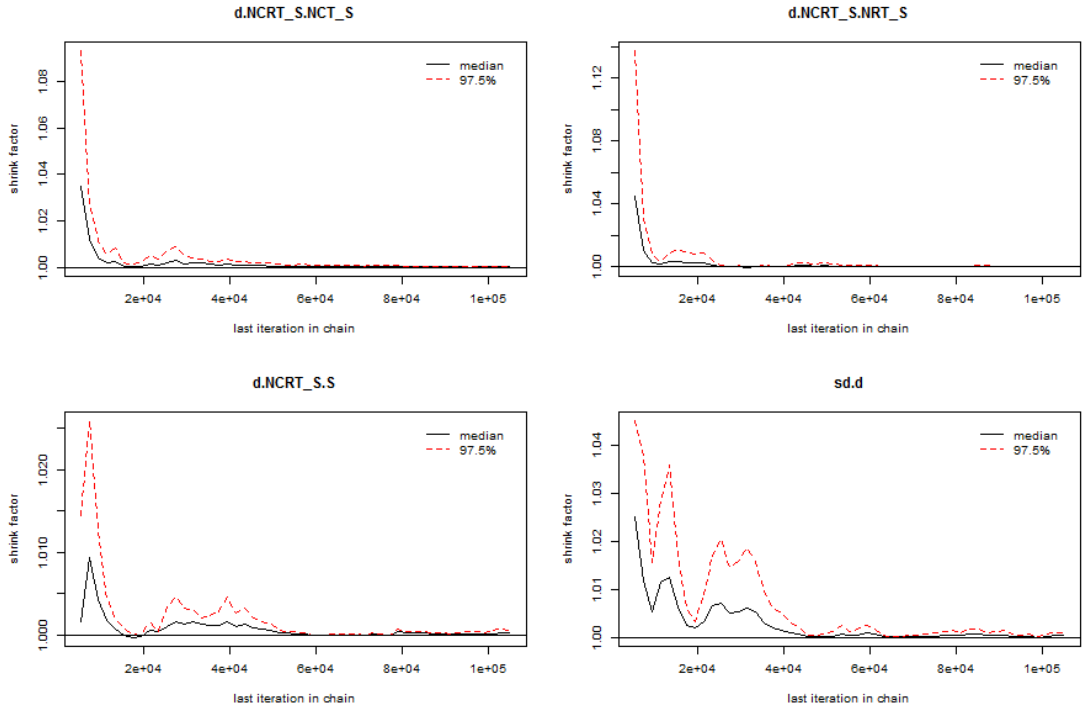

d.S.NRT_S
sd.d
median
97.5%
shrink factor
last iteration in chain
2e+04
4e+04
6e+04
8e+04
1e+05

f. DM


d.S.NCRT_S
d.S.NCT_S
median
97.5%
shrink factor
last iteration in chain
2e+04
4e+04
6e+04
8e+04
1e+05


d.S.NRT_S
sd.d
median
97.5%
shrink factor
last iteration in chain
2e+04
4e+04
6e+04
8e+04
1e+05

g. Subgroup analysis for pathological types: OS for patients with squamous cell carcinoma.



h. Subgroup analysis for pathological types: OS for patients with adenocarcinoma.

